A-A-53913
19 June 1989

COMMERCIAL ITEM DESCRIPTION

CHISELS, DENTAL (Carbon Steel)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-C-36804 which is cancelled.

Salient Characteristics:

This commercial item description covers thin bladed, carbon steel, dental chisels of Black's formula used in various steps of cavity preparation.

The dental chisels covered by this specification shall be as listed below:

No. 3
No. 14-15
No. 22-23
No. 24-25

Materials. The chisels shall be fabricated from water hardening carbon steel. The steel shall meet the requirements of ASTM A686 for Grade A (regular hardening), Class W1, 0.90 to 1.20 percent carbon, spheroidized annealed condition steel for: chemical composition, raw material hardness, macrostructure, surface decarburization, carbide form and distribution, hardness penetration and fracture grain size.

Hardness. The operative ends and shanks of the finished instrument shall have a Rockwell hardness of C56 to C62. The handle shall have a minimum Rockwell hardness of C20.

Style, design and dimensions. The chisels shall be one-piece, double ended, thin blade design. The No. 22-23 and No. 24-25 chisels shall be of Dr. Wedelstaedt's design modified by Dr. Ferrier. The No. 14-15 chisel shall be as modified by Dr. Ferrier. The No. 3 chisel shall be a dental, hoe type, enamel chisel. The style, design and dimensions of the chisels shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24287 for the applicable chisel. The chisels shall conform to the applicable Black's formula as specified in the drawing. The chisels shall be suitable for cutting, planing and cleaving enamel and dentin during dental operative procedures.

AMSC N/A FSC 6520

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Finish.

Operative ends. The beveled portion of the blade shall have a sharp cutting edge. Operative ends (blades) of chisels, including sharpened edges of the beveled portion, shall have a fine honed finish.

Handles and shanks. The chisel handles shall be straight and flat. Handles and shanks of chisels shall be plated as specified herein and shall have a bright finish. Handles shall have 40 ± 10 fillet or notch type serrations per inch on seven faces and one smooth face for identification marking. Depth of serrations shall be 0.002 to 0.004 inch.

Plating. The handles and shanks shall be chromium plated.

Corrosion resistance. The plated portions of the chisels shall be in accordance with the boil test requirements when tested as specified herein.

Identification marking. The chisels shall be permanently and legibly marked on the smooth face of the handle. The chisels shall be marked with the respective instrument number and Black's formula as specified herein. The respective instrument number shall be placed on the handle no less than 5 mm nor more than 12 mm from the junction of the handle and shank of the operative end. Chisels of foreign manufacture shall bear the name of the country of origin.

Workmanship. The operative end, handle and shank shall be free from coarse grind marks, cracks, burrs, feathers or any other defects which may detract from their appearance or impair their serviceability, proper functioning, or intended use.

Unit. Each (EA). One chisel, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality for raw material, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination and test characteristics. The unit of product for inspection, examination and test purposes shall be one chisel. Unless otherwise specified, sampling for examination shall be conducted at inpsection level II, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective) and sampling for tests (when applicable) shall be conducted at inspection level S-2 and acceptance number shall be zero (0) for all sample sizes.

Examination. Examination shall include, but shall not be limited to the following:

Defects

Style and design for the applicable type, class, and style chisel not as specified.
Dimensions not as specified.
Plating of the handle and shank incomplete or not continuous.
Handle and shank finish not as specified.
Handle not straight and flat.
Identification marking not present, properly located, legible, nor complete.
Chisels not free from dirt, polishing compound, or foreign matter.
Handle serrations not uniform or complete.

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Examination under magnification. The instruments shall be examined under four power (4x) magnification for the following defects:

Defects

Cutting edge of operative end rounded or not sharp.
Cutting edge not free from nicks, burrs, feathers, indentations or other irregularities that affect cutting performance.
Operatiive end finish not as specified.
Plating of handles and shank not free from flaking, cracks or blisters.

Tests.

Chemical analysis. In the absence of certificates of quality, the chemical analysis of the instrument shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of a conflict, method 111 shall govern.

Hardness. The finished instrument hardness shall be determined in accordance with ASTM E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Boil test. The chisels shall be subjected to the boil test as specified in ASTM F1089.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine of the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maxium extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-469-3499	CHISEL,DENTAL, BLACK'S FORMULA 11-1/2, 15, 3; No. 22-23 Carbon Steel
6520-00-469-3500	CHISEL, DENTAL, BLACK'S FORMULA 15, 15, 3; No. 24-25 Carbon Steel

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6520-00-469-3502	CHISEL, DENTAL, BLACK'S FORMULA 18, 10, 16; No. 14-15 Carbon Steel
6520-01-074-1814	CHISEL, DENTAL, BLACK'S FORMULA 6-1/2, 2-1/2, 9; NO. 3 Carbon Steel

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

05404

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53913	CHISELS, DENTAL (Carbon Steel)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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